



Effect of refining processes on antioxidant capacity, total contents of phenolics and carotenoids in palm oils

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ABSTRACT

Antioxidant capacity (AC), total phenolic content (TPC) and total carotenoid content (TCC) in palm oils at various stages of the refining process from two technological modes were determined. The obtained mean FRAP and DPPH values for the methanolic extracts of palm oils from mode 1 (19.5–102.8 $\mu\text{mol TE}/100\text{ g}$ and 18.8–103.0 $\mu\text{mol TE}/100\text{ g}$) were lower than for oils from mode 2 (25.6–134.8 $\mu\text{mol TE}/100\text{ g}$ and 25.4–135.4 $\mu\text{mol TE}/100\text{ g}$). The total phenolics (4.1–12.4 mg GA/100 g) and carotenoids (0.18–45.8 mg/100 g) in the studied oils were correlated with their antioxidant capacities determined by FRAP and DPPH methods ($r = 0.6623$ – 0.9878). During the refining process, for both technological modes resulted in a loss of AC by 80%, TPC by 26–55% and TCC by 99%. The bleaching step caused the highest losses of AC as determined by FRAP 41% and 46%, DPPH by 43% and 48%, while TPC loss was 45% and 23% and loss of carotenoids was 49% and 56%, in mode 1 and mode 2, respectively.

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1. Introduction

Palm fruits and crude palm oils (CPO) are known as rich sources of antioxidants (tocopherols, tocotrienols, carotenoids, phytosterols, phenolic compounds) and other phytonutrients essential for good human health (Edem, 2002; Rossi, Gianazza, Alamprese, & Stanga, 2001; Sambanthamurthi, Sundram, & Tan, 2000). However, CPO must be refined by a physical refining method, which includes three major processes: degumming, bleaching and deodorization to remove undesirable components for other applications (Manan, Siang, & Mustapa, 2009).

Various analytical procedures such as: Soxhlet extraction (Chan & Ismail, 2009; Sanagi, See, Ibrahim, & Naim, 2005), pressurised liquid extraction (PLE) (Sanagi et al., 2005) and solvent extraction using mixtures of methanol and water in different proportions (Biglari, Alkarkhi, & Easa, 2008; Farombi & Britton, 1999; Mansouri, Embarek, Kokkalou, & Kefalas, 2005), 60% ethanol in water (Neo, Ariffin, Tan, & Tan, 2008) were used for isolation of antioxidants from palm fruits and oils. Supercritical carbon dioxide (SC-CO₂) extraction was applied to recover the water-soluble compounds from fresh palm pressed fibre (Nang, May, Ngan, & Hock, 2007) and β -carotene from crude palm oil (Davarnejad, Kassim, Zainal, & Sata, 2008). Also, phenolic compounds were extracted in soluble free, insoluble-bound and esterified forms from palm fruits by solvent extraction and base hydrolysis (Neo, Ariffin, Tan, & Tan, 2010; Neo et al., 2008). Generally, the spectrophotometric Folin–Ciocal-

teu's procedure has been used for determination of total phenolic content (TPC = 2.49–8397 mg gallic acid/100 g) in palm fruits (Balasundram, Ai, Sambanthamurthi, Sundram, & Samman, 2005; Biglari et al., 2008; Gorret et al., 2004; Mansouri et al., 2005; Neo et al., 2008, 2010).

Moreover, the total content of flavonoids (TFC = 1.62–753 mg catechin equivalents/100 g) and carotenoids (TCC = 9.16–600 mg/100 g) in palm fruits and oils were determined spectrophotometrically (Biglari et al., 2008; Davarnejad et al., 2008; Farombi and Britton, 1999; Fernandez, Shier, & Watkins, 2000; Neo et al., 2008; Rossi et al., 2001; Sanagi et al., 2005; Schroeder, Becker, & Skibsted, 2006). However, for separation, identification, and quantification of the individual phenolic and carotenoid compounds, chromatographic techniques were adopted (Darnoko, Cheryan, Moros, Jerrel, & Perkins, 2000; Farombi and Britton, 1999; Mansouri et al., 2005; Mortensen, 2005; Nang et al., 2007; Neo et al., 2010; Ng & Tan, 1988; Ping, 2006). Polyphenols, carotenoids, tocopherols, and other antioxidants present in palm fruits and oils exhibit antioxidant capacity (AC), causing the reduction of the oxidative stress attributed to chronic diseases, heart diseases, neurodegenerative diseases, cancers, diabetes (Biglari et al., 2008; Neo et al., 2008). Different spectrophotometric methods, including: 2,2'-diphenyl-1-picrylhydrazyl (DPPH) (Balasundram et al., 2005; Chan and Ismail, 2009; Chiou, Kalogeropoulos, Salta, Efstathiou, & Andrikopoulos, 2009; Mansouri et al., 2005; Neo et al., 2008, 2010; Wu & Ng, 2007), ferric reducing antioxidant power (FRAP) (Biglari et al., 2008; Neo et al., 2010), 2,2'-azinobis (3-ethylbenzothiazoline-6-sulphonic acid) radical cation (ABTS⁺) (Biglari et al., 2008; Neo et al., 2010), and β -carotene bleaching (Chan and Ismail, 2009)

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